2-Channel Time Interleaved ADC

Team A03 Aevulog

About Team

Name	Discord name	Affiliation (experience)	Role
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Project Information

- Two 8-bit SAR ADCs are interleaved to achieve 50 MS/s throughput on GF180MCU, each channel sampling at 25 MS/s
- Each SAR ADC uses a double-tail comparator and a split-array CDAC, for lower power and faster operation
- Digital background calibration corrects offset and gain mismatch between channels without interrupting operation
- Review Date: 7/4/2026
- Track A
- Review Issue: Issue #23
- Review Repository: chipathon-2026-ti-adc

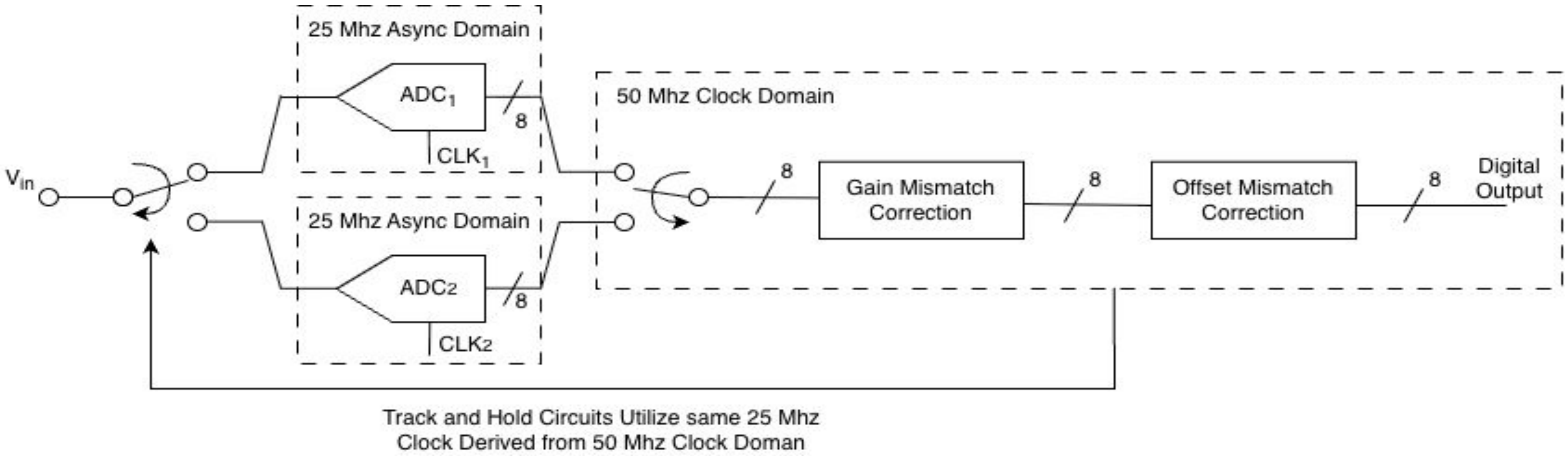
Design Objective

- The circuit enables faster sampling utilizing a SAR ADCs, doubling the sampling rate of an ADC
- It is intended to be used as a foundational ADC that can be used whenever a digital block needs to interact with analog voltages.

Target Specifications

Parameters	Specifications
Area	Block B
Sampling Rate	50 Ms / sec
Bits	8 bits
ENOB / SFDR	7.2 bits / 60 dBc
Input Range	0.0V - 3.3V

System Overview

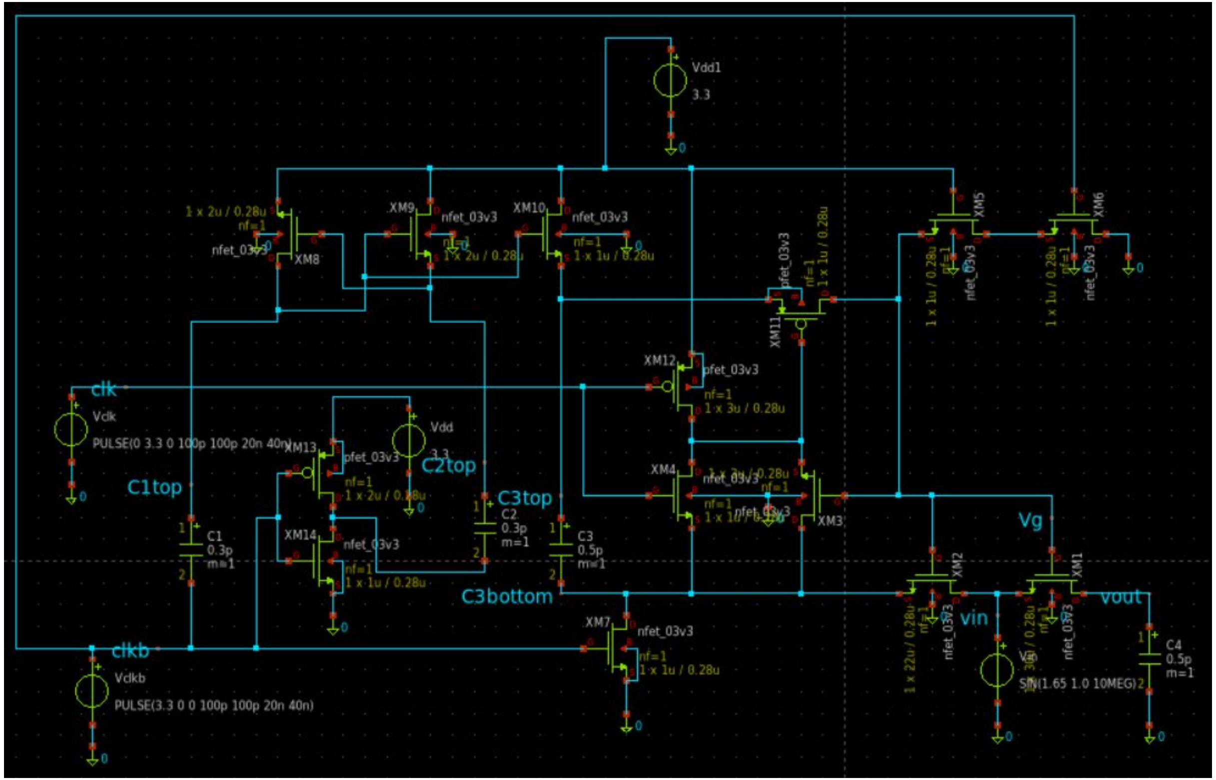


Track and Hold Circuit

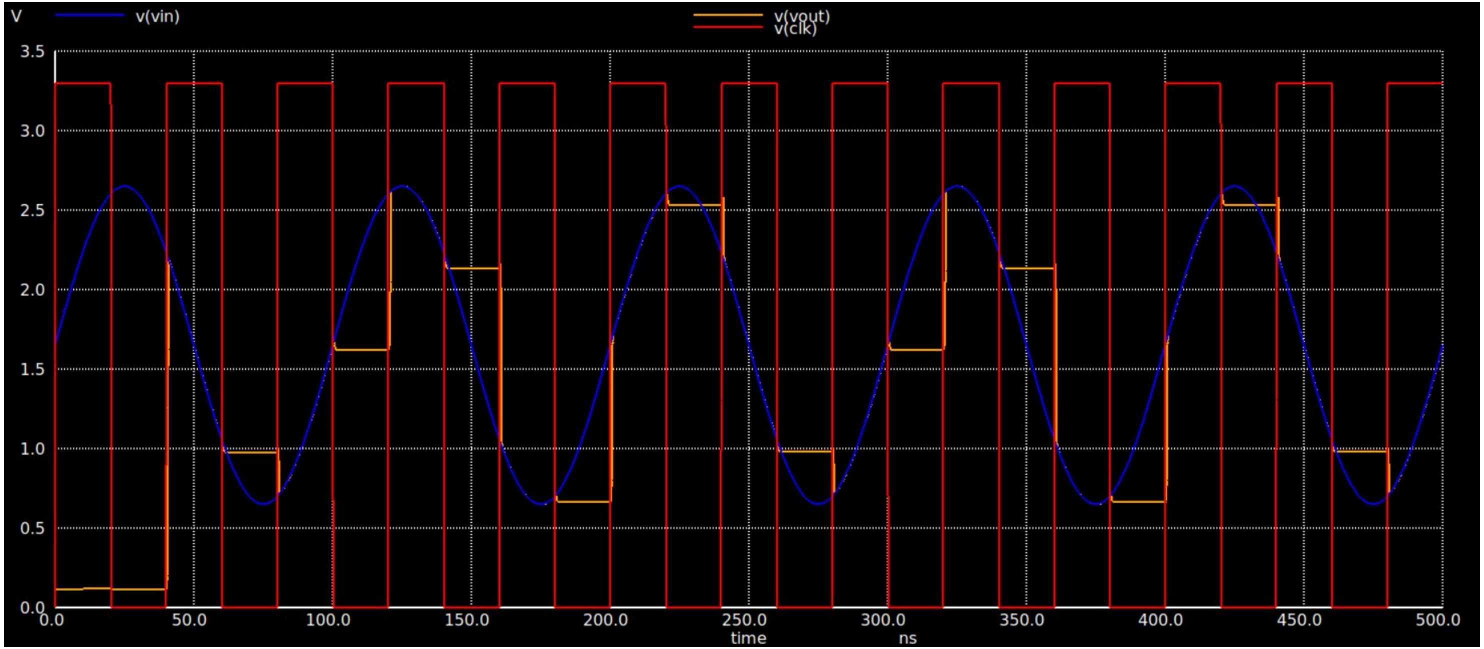
Purpose:
Hold input voltage steady while ADC makes conversion.

Inputs: VIN, CLK

Outputs: VOUT



- The bootstrapped switch first precharges a capacitor to Vdd while the sampling nmos is off.
- During track, that capacitor is floated and stacked on top of the input signal, driving the sampling gate to approximately: Vg = Vin + Vdd



Double-Tailed Comparator

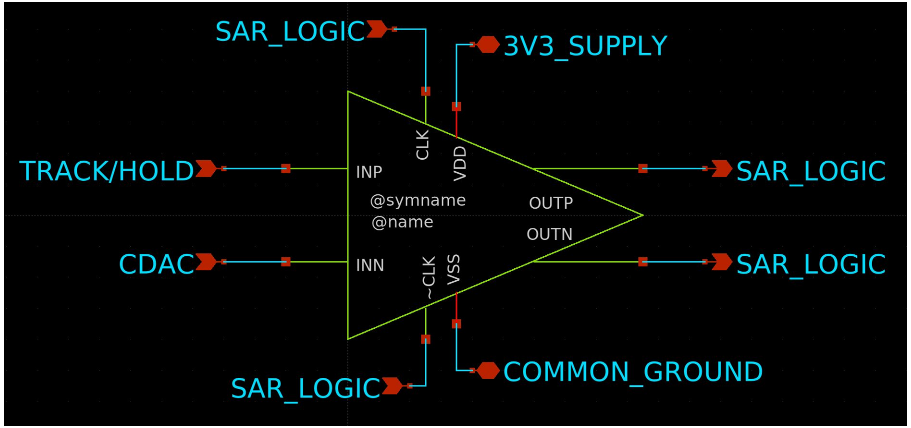
Purpose:

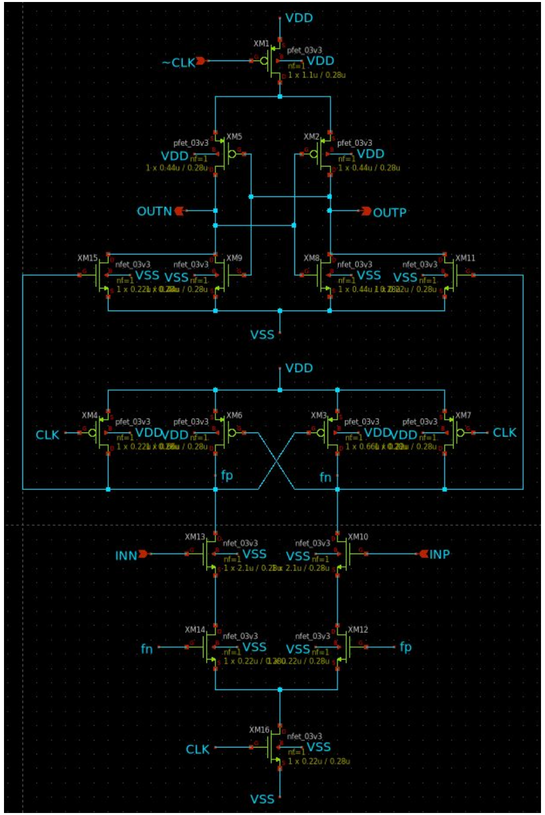
Compare the analog input from T/H to each consecutive binary-searched voltage

Inputs: compare-clk (200MS/s), INP & INN (0V-3.3V)

Outputs: OUTP, OUTN (0V-3.3V)

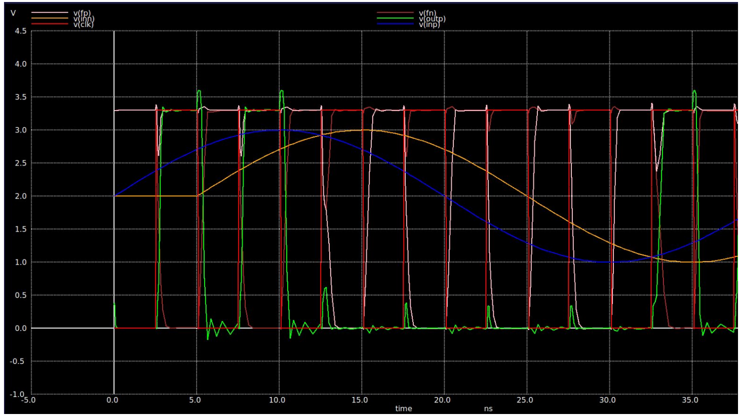
Interfaces with other blocks:



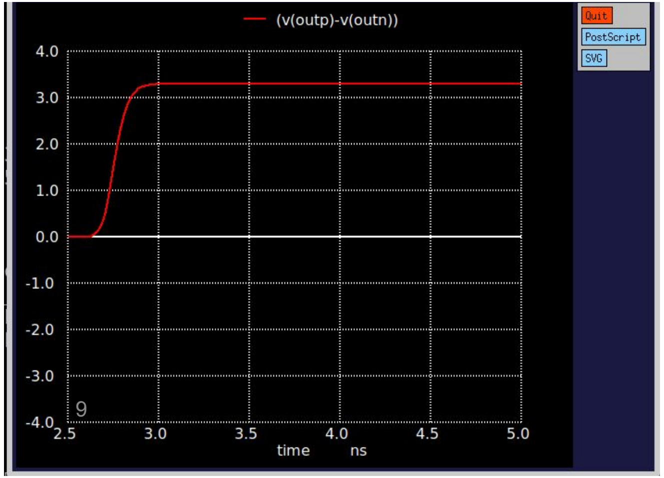


Design Decisions

General Operation

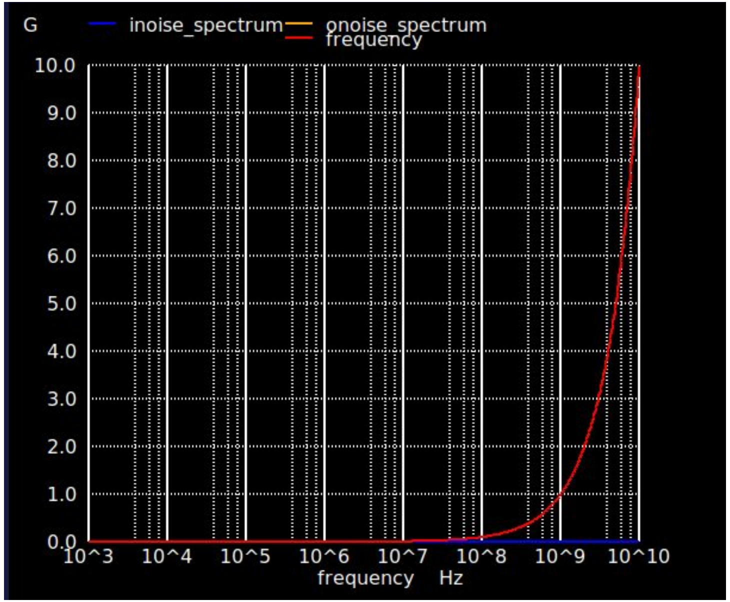


Metastability



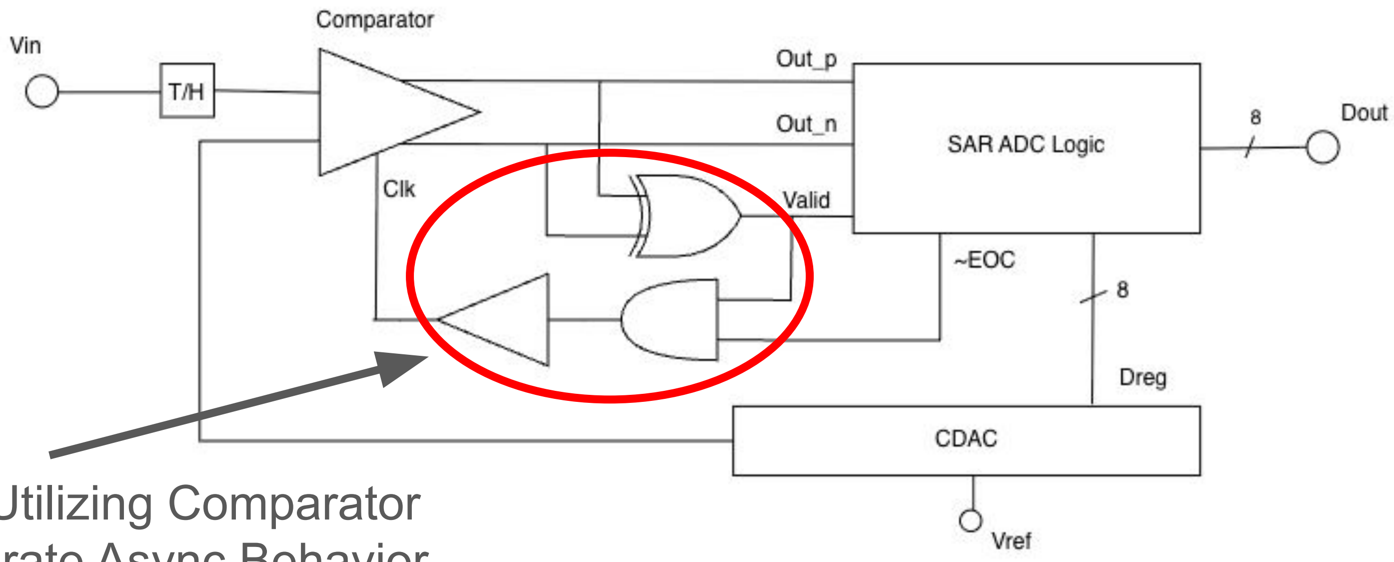
Tau_reneg:
1.2 ns
2 time constants of growth

Gain of Noise across frequencies

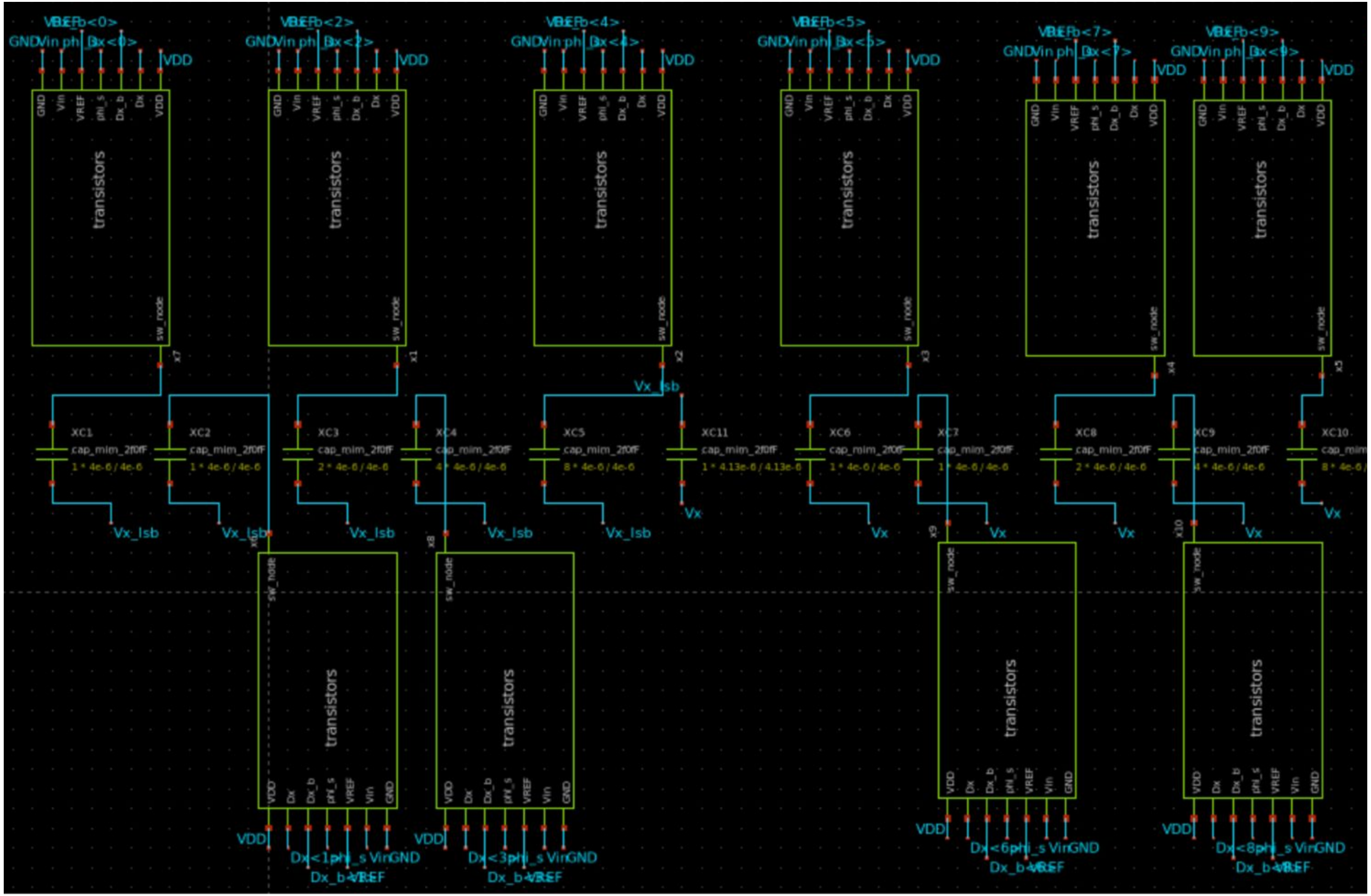


SAR ADC Logic

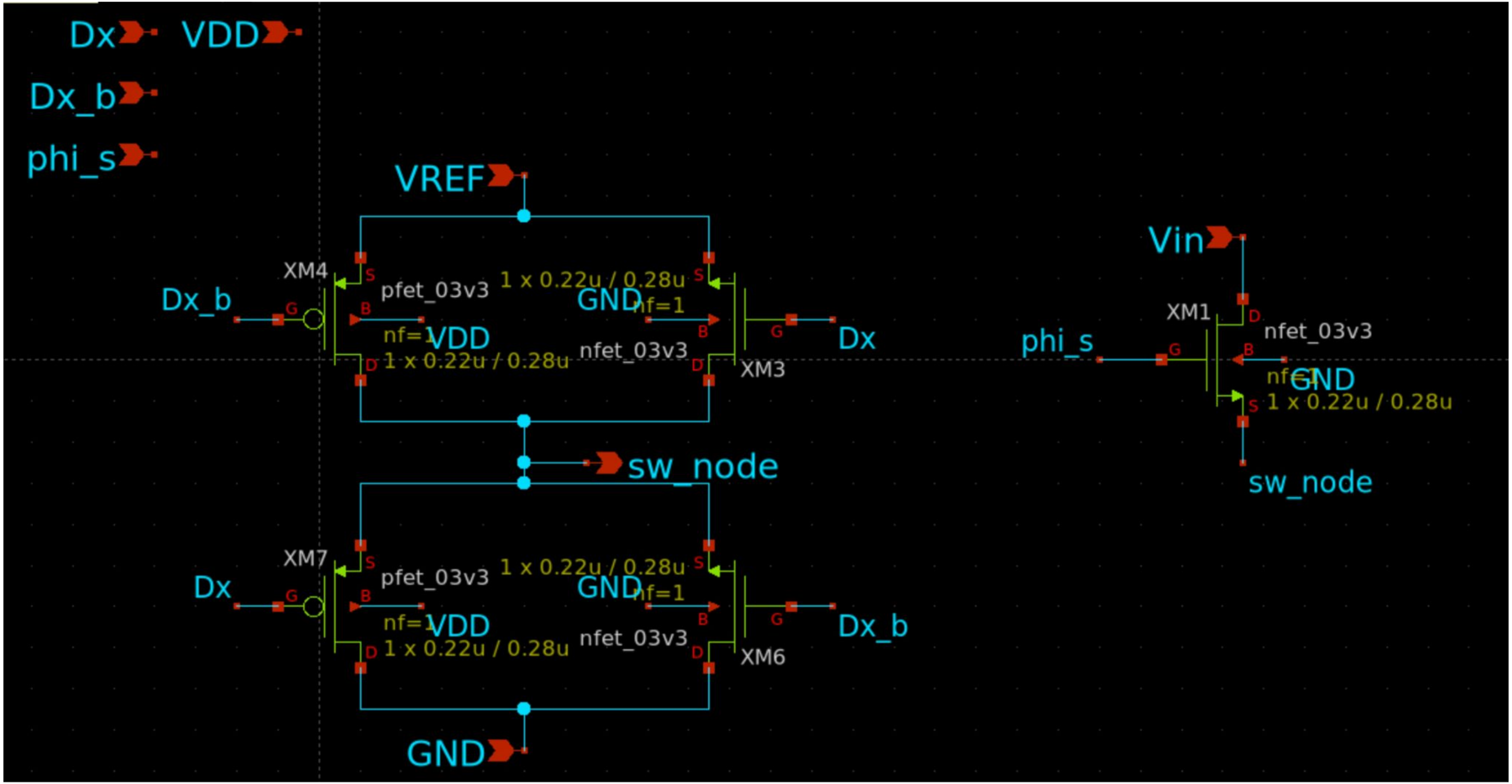
Purpose: Execute binary search logic of the SAR ADC
Inputs: Comparator Differential Outputs, Outputs: Dout, Dreg, Valid, EOC

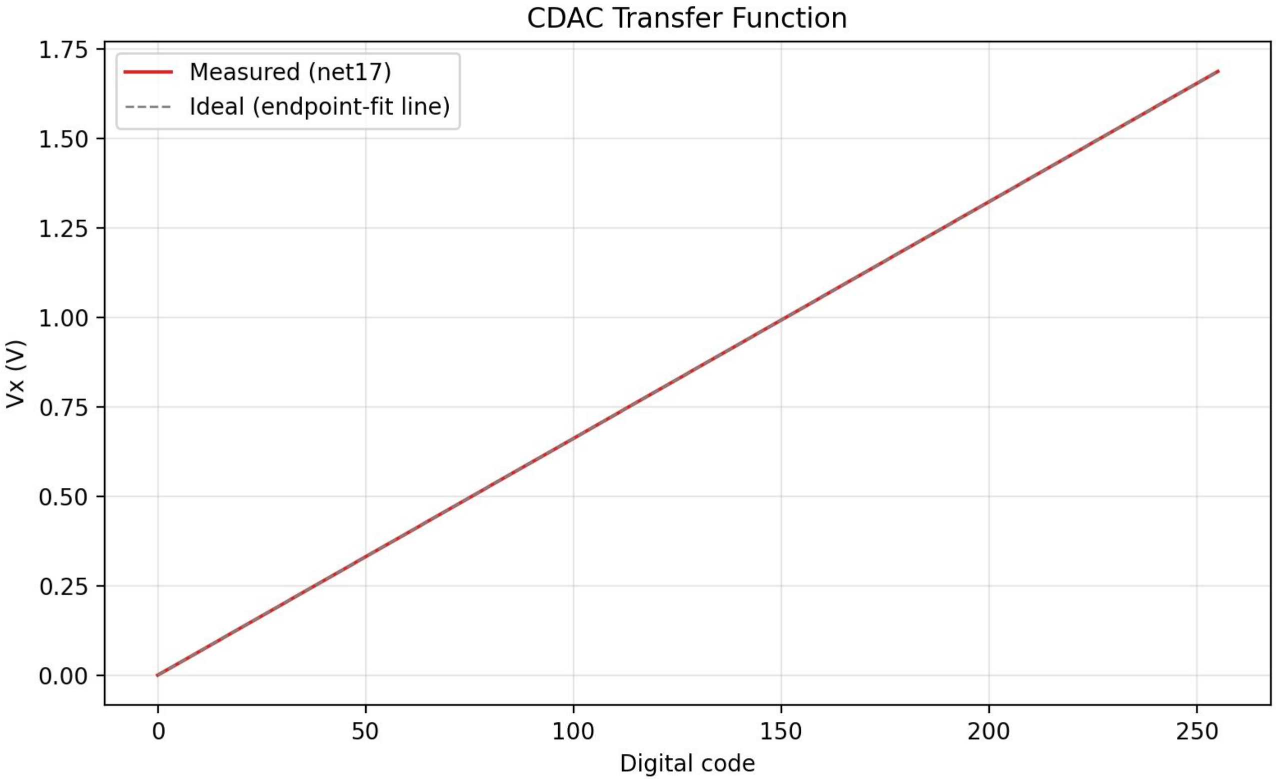


Custom Clock Utilizing Comparator Output to Generate Async Behavior



transistors
XC1
XC2
XC3
XC4
XC5
XC6
XC7
XC8
XC9
XC10
XC11
cap_mim_2f0fF
Vx_lsb
Vx
VDD


Dx
VDD
Dx_b
phi_s
VREF
XM4
pfet_03v3
1 x 0.22u / 0.28u
nf=1
VDD
GND
nf=1
nfet_03v3
XM3
Dx_b
Dx
Vin
XM1
nfet_03v3
phi_s
nf=1
GND
1 x 0.22u / 0.28u
sw_node
sw_node
XM7
pfet_03v3
1 x 0.22u / 0.28u
nf=1
VDD
GND
nf=1
nfet_03v3
XM6
Dx
Dx_b
GND

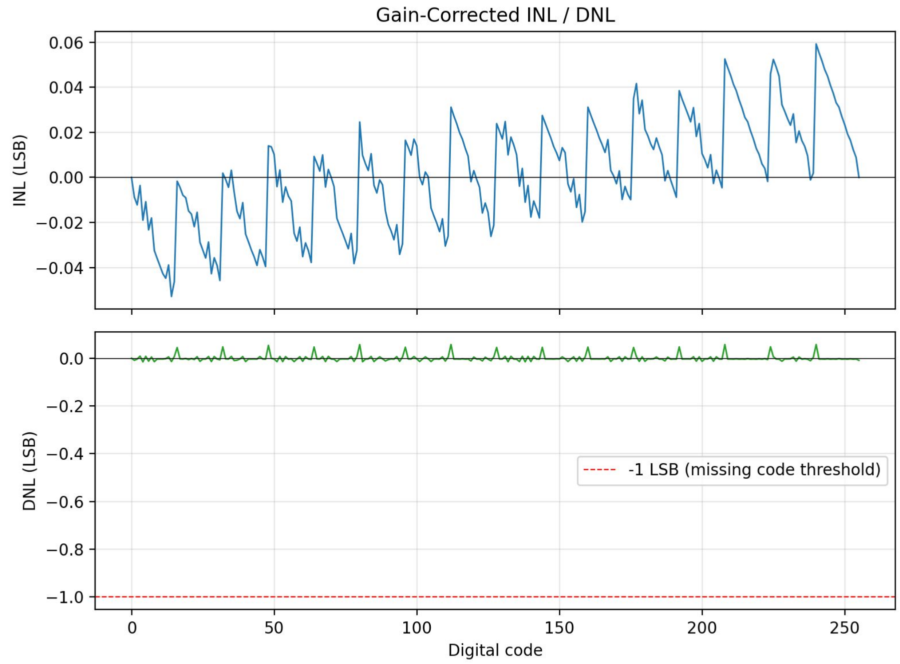


- Monotonic ramp across all 256 codes, full-scale sweep
- Zero missing codes (DNL never crosses -1 LSB)

INL/DNL (gain-corrected)

- DNL: +0.057 / −0.015 LSB
- INL: +0.059 / −0.053 LSB
- Gain error: −5.88% (systematic, calibratable)

Sub-0.06 LSB INL/DNL show the array is highly linear; the -5.88% gain error is a fixed, correctable offset

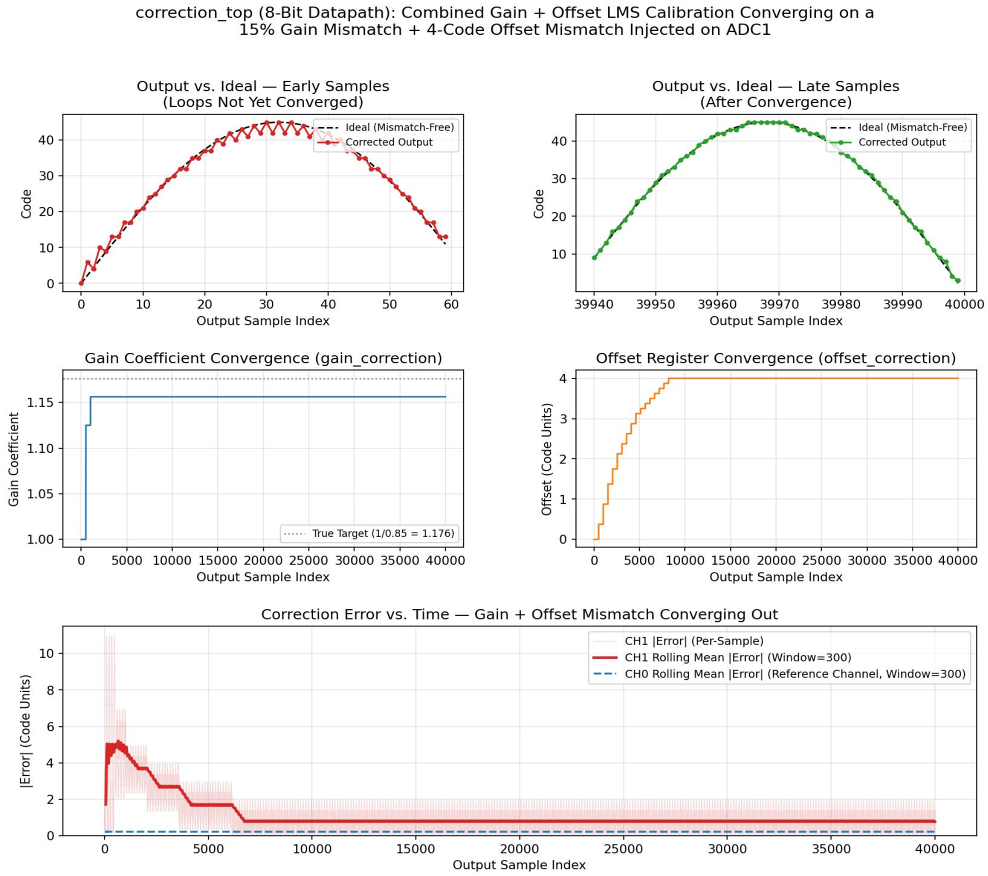


Gain Mismatch Digital Correction

- Dynamically corrects inter-channel gain mismatch in a time-interleaved architecture by treating CH0 as the reference and scaling CH1 to match its signal envelope.
- Accumulates the absolute values of both ADC channels over a configurable sample window to extract the average signal magnitude and calculate the instantaneous gain error.
- Adjusts a fixed-point gain coefficient continuously using an LMS-style feedback loop, utilizing a computationally cheap, right-shifted step size (mu) instead of complex division.
- Features an activity-monitoring threshold to freeze the calibration loop during DC or low-signal periods, preventing coefficient divergence, alongside a 2-stage pipelined datapath with saturation logic to prevent overflow.

Offset Mismatch Digital Correction

- Compensates for DC offset mismatch between time-interleaved channels by dynamically aligning the baseline of CH1 to the CH0 reference.
- Averages incoming signed samples over a configurable, power-of-two window to filter out AC signal components and isolate the instantaneous DC error.
- Updates the offset correction value using a computationally light, shift-based step size (mu) into a high-precision register, preserving fractional bits to ensure stable, fine-grained convergence.
- Subtracts the rounded, integer-converted offset from CH1 in a pipelined stage, utilizing built-in saturation logic to protect the final output against numerical overflow or wrapping.



Digital Correction

- Parameterizable RTL for both gain correction and offset correction finished
- Final parameters need to be determined via simulations
- Sanity Testbenches show promising results of being able to remove injected errors
  - More thorough verification needs to be completed
- Netlist and sanity testbenches are viewable on our Github Repository

Design Checklist

Digital Tasks	Status
System Architecture and Design Specs	Completed
RTL Design	Completed
RTL Verification	In Progress
Floor Plan + Layout	In Progress

Analog Tasks	Status
Schematic Design	Completed
Spice Simulations	Completed
Layout Design	In Progress
Parasitic Simulations	In Progress
Mixed-Signal Integration	Not started

Design Assumptions

- Supply Voltages: 3.3V
- Temperature Range: 25ºC - 125ºC
- Input Range 0.0V - 3.3V
- External Circuitry: Input Voltage to be measured

Questions

1. Should we change comparator topology to be able to input negative voltages?
2. Do you have any recommendations for how to thoroughly verify all of the digital blocks?
3. After simulating and verifying every individual block, what is the best way to integrate them all together?
4. What should we as a team focus on to ensure that we get to Tapeout / what are the most pressing challenged we aren't considering?

Resources

- J. Elbornsson, F. Gustafsson and J. . -E. Eklund, "Blind adaptive equalization of mismatch errors in a time-interleaved A/D converter system," in *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 51, no. 1, pp. 151-158, Jan. 2004
- Daihong Fu, K. C. Dyer, S. H. Lewis and P. J. Hurst, "A digital background calibration technique for time-interleaved analog-to-digital converters," in *IEEE Journal of Solid-State Circuits*, vol. 33, no. 12, pp. 1904-1911, Dec. 1998
- Splitarray origin: B. Ginsburg & A. Chandrakasan, "500-MS/s 5-bit ADC in 65-nm CMOS," *JSSC* 2006
- Hariprasath et al., "Merged capacitor switching based SAR ADC," *Electronics Letters* 2010
- "Capacitor mismatch effects in SC circuits," *ISCAS* 1990
- *S. Babayan-Mashhadi and R. Lotfi, "Analysis and Design of a Low-Voltage Low-Power Double-Tail Comparator," in IEEE Transactions on Very Large Scale Integration (VLSI)*